



An Introduction to Using Data

Useful Formulas and Where to Use Them

Please do not copy without permission. © ALX 2024.

“One cannot escape the feeling that these **mathematical formulas have an independent existence and an intelligence of their own**, that they are wiser than we are, wiser even than their discoverers ...”

– Heinrich Hertz

Why should I learn this?

Trying to organize and analyze spreadsheets can feel daunting if you're unfamiliar with **formulas** and when to use them.

01. Increased efficiency

- Formulas allow for **more efficient** data analysis and complex calculations by simplifying the way we process data.
- **Result:** A cleaner, easier-to-read spreadsheet. Useful for presentations.

02. Increased productivity

- Formulas allow us to **capture and store working solutions**.
- **Result:** A reusable solution with perfect accuracy every time, and get work done faster.

03. Makes you stand out

- Being good at formulas can **add** to your **value**.
- **Result:** Being comfortable with formulas will make you a vital asset to companies that rely on spreadsheets to make data-driven decisions.

Data overview

The gender pay gap dataset contains 20 rows.
We will focus on the following columns:

A. Age

The age of an employee.

B. Base pay

The annual salary of an employee, excluding any other compensation.

C. Bonus

Money is given to employees beyond their base pay every year.



The dataset

	A	B	C	D	E	F
1	Gender	JobTitle	Age	Education	BasePay	Bonus
2	Female	Graphic Designer	18	College	42363.00	9938.00
3	Male	Software Engineer	21	College	108476.00	11128.00
4	Female	Warehouse Associate	19	PhD	90208.00	9268.00
5	Male	Software Engineer	20	Masters	108080.00	10154.00
6	Male	Graphic Designer	26	Masters	99464.00	9319.00
7	Female	IT	20	PhD	70890.00	10126.00
8	Female	Graphic Designer	20	College	67585.00	10541.00
9	Male	Software Engineer	18	PhD	97523.00	10240.00
10	Female	Graphic Designer	33	High School	112976.00	9836.00
11	Female	Sales Associate	35	College	106524.00	9941.00
12	Male	Graphic Designer	24	PhD	102261.00	10212.00
13	Female	Driver	18	College	62759.00	10124.00
14	Female	Financial Analyst	19	College	84007.00	8990.00

Relative and absolute cell references

A reference is a cell address that identifies a cell or range of cells by referring to the **column letter** and **row number** of the cell or cells.

Ranges are referenced using **two cell references** separated by a **colon** and can span multiple rows and columns.

Cell referencing is important because it:

- Allows formulas to **automatically update** when specific **cell values change**.
- Helps **formulas update** when cells are **copied or moved**.

We can use **relative** references, **absolute** references, or a mixture of the two.

	A	B	C	D	E	F
1	A1				D1:F1	
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						

Relative and absolute cell references

Relative references:

- Relative references **change when copied** across multiple cells based on the relative position of rows and columns.
- For example, if the formula **=A1+B1** is copied from row 1 to row 2, the formula in row 2 will become **=A2+B2**.
- All cell references are relative references by **default**.

01. Enter the formula **=C2*2** in cell D2.

02. Replicate the formula to other rows by dragging down the fill handle.

D2	fx =C2*2						
	A	B	C	D	E	F	G
1	Gender	JobTitle	Age	Relative	Education	BasePay	Bonus
2	Female	Graphic Designer	18	36	College	42363.00	9938.00
3	Male	Software Engineer	21		College	108476.00	11128.00
4	Female	Warehouse Associate	19		PhD	90208.00	9268.00

D10	fx =C10*2						
	A	B	C	D	E	F	G
1	Gender	JobTitle	Age	Relative	Education	BasePay	Bonus
2	Female	Graphic Designer	18	36	College	42363.00	9938.00
3	Male	Software Engineer	21	42	College	108476.00	11128.00
4	Female	Warehouse Associate	19	38	PhD	90208.00	9268.00
5	Male	Software Engineer	20	40	Masters	108080.00	10154.00
6	Male	Graphic Designer	26	52	Masters	99464.00	9319.00
7	Female	IT	20	40	PhD	70890.00	10126.00
8	Female	Graphic Designer	20	40	College	67585.00	10541.00
9	Male	Software Engineer	18	36	PhD	97523.00	10240.00
10	Female	Graphic Designer	33	66	High School	112976.00	9836.00
11	Female	Sales Associate	35	70	College	106524.00	9941.00
12	Male	Graphic Designer	24	48	PhD	102261.00	10212.00
13	Female	Driver	18	36	College	62759.00	10124.00
14	Female	Financial Analyst	19	38	College	84007.00	8990.00
15	Female	Warehouse Associate	30	60	Masters	86220.00	9583.00

Relative and absolute cell references

Absolute references:

- Absolute references **keep a row and/or column constant** by using the **dollar sign (\$)** before the row number and/or column letter. For example, if the formula **=A\$1+B1** is copied from row 1 to row 2, the formula in row 2 will become **=A\$1+B2**.
- How to use:
 - \$A\$1** – The column and row do not change.
 - \$A1** – The column does not change.
 - A\$1** – The row does not change.

01. Enter the formula **=C\$2*2** in cell D2.

02. Replicate the formula to other rows by dragging down the fill handle.

D2				fx	=C\$2*2		
	A	B	C	D	E	F	G
1	Gender	JobTitle	Age	Absolute	Education	BasePay	Bonus
2	Female	Graphic Designer	18	36	College	42363.00	9938.00
3	Male	Software Engineer	21		College	108476.00	11128.00
4	Female	Warehouse Associate	19		PhD	90208.00	9268.00

D10				fx	=C\$2*2		
	A	B	C	D	E	F	G
1	Gender	JobTitle	Age	Absolute	Education	BasePay	Bonus
2	Female	Graphic Designer	18	36	College	42363.00	9938.00
3	Male	Software Engineer	21	36	College	108476.00	11128.00
4	Female	Warehouse Associate	19	36	PhD	90208.00	9268.00
5	Male	Software Engineer	20	36	Masters	108080.00	10154.00
6	Male	Graphic Designer	26	36	Masters	99464.00	9319.00
7	Female	IT	20	36	PhD	70890.00	10126.00
8	Female	Graphic Designer	20	36	College	67585.00	10541.00
9	Male	Software Engineer	18	36	PhD	97523.00	10240.00
10	Female	Graphic Designer	33	36	High School	112976.00	9836.00
11	Female	Sales Associate	35	36	College	106524.00	9941.00
12	Male	Graphic Designer	24	36	PhD	102261.00	10212.00
13	Female	Driver	18	36	College	62759.00	10124.00
14	Female	Financial Analyst	19	36	College	84007.00	8990.00
15	Female	Warehouse Associate	30	36	Masters	86220.00	9583.00

The AVERAGE and AVERAGEA functions

Used to find the **arithmetic mean** for a list of arguments, i.e., the sum of the values in the arguments divided by the number of arguments. They differ in how they evaluate boolean and text values.

```
=AVERAGE(value1, [value2, ...])
=AVERAGEA(value1, [value2, ...])
```

Arguments can take the form of:

- A list of numbers with the maximum number of entries allowed being 30:

```
=AVERAGE(13,24,107) or =AVERAGEA(13,24,107)
```

- Cell references:

```
=AVERAGE(B2, H17, G9) or =AVERAGEA(B2, H17, G9)
```

- Range of cells:

```
=AVERAGE(B2:B24) or =AVERAGEA(B2:B24)
```

E21						
E22						
	A	B	C	D	E	F
1	Gender	JobTitle	Age	Education	BasePay	Bonus
2	Female	Graphic Designer	18	College	42363.00	9938.00
3	Male	Software Engineer	21	College	108476.00	11128.00
4	Female	Warehouse Associate	19	PhD	90208.00	9268.00
5	Male	Software Engineer	20	Masters	108080.00	10154.00
6	Male	Graphic Designer	26	Masters	99464.00	9319.00
7	Female	IT	20	PhD	70890.00	10126.00
8	Female	Graphic Designer	20	College	67585.00	10541.00
9	Male	Software Engineer	18	PhD	97523.00	10240.00
10	Female	Graphic Designer	33	High School	112976.00	9836.00
11	Female	Sales Associate	35	College	106524.00	9941.00
12	Male	Graphic Designer	24	PhD	102281.00	10212.00
13	Female	Driver	18	College	62759.00	10124.00
14	Female	Financial Analyst	19	College	84007.00	8990.00
15	Female	Warehouse Associate	30	Masters	86220.00	9583.00
16	Female	Warehouse Associate	35	PhD	95584.00	9745.00
17	Female	Marketing Associate	27	PhD	73357.00	10334.00
18	Female	Financial Analyst	23	PhD	88422.00	10768.00
19	Female	Warehouse Associate	24	College	99545.00	9949.00
20	Male	Sales Associate	21	High School	90386.00	9461.00
21					Salary AVERAGE	88770.00
22					Salary AVERAGEA	88770.00

The AVERAGE and AVERAGEA functions

F2	fx =E2-E21					
	A	B	C	D	E	F
1	Gender	JobTitle	Age	Education	BasePay	Av. BasPay Disparity
2	Female	Graphic Designer	18	College	42363.00	-46407.00
3	Male	Software Engineer	21	College	108476.00	19706.00
4	Female	Warehouse Associate	19	PhD	90208.00	1438.00
5	Male	Software Engineer	20	Masters	108080.00	19310.00
6	Male	Graphic Designer	26	Masters	99464.00	10694.00
7	Female	IT	20	PhD	70890.00	-17880.00
8	Female	Graphic Designer	20	College	67585.00	-21185.00
9	Male	Software Engineer	18	PhD	97523.00	-21185.00
10	Female	Graphic Designer	33	High School	112976.00	24206.00
11	Female	Sales Associate	35	College	106524.00	17754.00
12	Male	Graphic Designer	24	PhD	102261.00	13491.00
13	Female	Driver	18	College	62759.00	-26011.00
14	Female	Financial Analyst	19	College	84007.00	-4763.00
15	Female	Warehouse Associate	30	Masters	86220.00	-2550.00
16	Female	Warehouse Associate	35	PhD	95584.00	6814.00
17	Female	Marketing Associate	27	PhD	73357.00	-15413.00
18	Female	Financial Analyst	23	PhD	88422.00	-348.00
19	Female	Warehouse Associate	24	College	99545.00	10775.00
20	Male	Sales Associate	21	High School	90386.00	1616.00
21				Salary AVERAGE	88770.00	
22				Salary AVERAGEA	88770.00	

Example use:

Determine if there is a **disparity** between employee annual **base pay** based on **gender**.

01.

Determine the average of **BasePay** for all employees.

02.

Subtract the calculated average salary from **BasePay** for each employee.



Insight

Our dataset reveals that **only female employees** are paid a salary that is **below the average rate**. This is irrespective of their education level, age, or job title.

The AVERAGE and AVERAGEA functions

01. **Floating-point** numbers will always return a **floating-point** result.
02. **Integer** numbers can produce either a **floating-point** or **integer** number.
03. A **mix** of floating-point and integer data types will result in a **floating-point**.

female	Driver	18	College	62759.00	-2
	02. Financial Analyst	19	College	84007.00	-
female	Warehouse Associate	30	Masters	86220.00	-
female	Warehouse Associate	35	PhD	95584.00	
female	Marketing Associate	27	PhD	73357.00	-1
female	Financial Analyst	23	PhD	88422.00	
female	Warehouse Associate	24	College	99545.00	1
male	Sales Associate	21	High School	90386.00	
				Salary AVERAGE	88770.00
				Salary AVERAGEA	88770.00

female	Driver	18	College	62759.50	
	01. Financial Analyst	19	College	84007.50	
female	Warehouse Associate	30	Masters	86220.50	
female	Warehouse Associate	35	PhD	95584.50	
female	Marketing Associate	27	PhD	73357.50	
female	Financial Analyst	23	PhD	88422.50	
female	Warehouse Associate	24	College	99545.50	
male	Sales Associate	21	High School	90386.50	
				Salary AVERAGE	88770.50
				Salary AVERAGEA	88770.50

female	Driver	18	College	62759.00	-
	03. Financial Analyst	19	College	84007.00	
female	Warehouse Associate	30	Masters	86220.00	
female	Warehouse Associate	35	PhD	95584.33	
female	Marketing Associate	27	PhD	73357.40	-
female	Financial Analyst	23	PhD	88422.50	
female	Warehouse Associate	24	College	99545.00	
male	Sales Associate	21	High School	90386.33	
				Salary AVERAGE	88770.08
				Salary AVERAGEA	88770.08

The AVERAGE and AVERAGEA functions

04. Empty cells will be ignored.

AVERAGE will ignore cells with **text**, including cells with **boolean** values (True and False).

AVERAGEA will translate all **text to zero**, **True** to **1**, and **False** to **0**.

05. An **error** will be returned if an argument contains an error.

13	Female	Driver	18	College	62759.00	
14	Male	Financial Analyst	19	College	TRUE	
15	Female	Warehouse Associate	30	Masters	86220.00	
16	Female	Warehouse Associate	35	PhD		
17	Female	Marketing Associate	27	PhD	73357.00	
18	Female	Financial Analyst	23	PhD	88422.00	
19	Female	Warehouse Associate	24	College	Analysis	
20	Male	Sales Associate	21	High School	90386.00	
21						
22						
					Salary AVERAGE	87968.38
					Salary AVERAGEA	78194.17

13	Female	Driver	18	College	62759.00	
14	Male	Financial Analyst	19	College	84007.00	
15	Female	Warehouse Associate	30	Masters	86220.00	
16	Female	Warehouse Associate	35	PhD	95584.00	
17	Female	Marketing Associate	27	PhD	73357.00	
18	Female	Financial Analyst	23	PhD	#NAME?	
19	Female	Warehouse Associate	24	College	99545.00	
20	Male	Sales Associate	21	High School	90386.00	
21						
22						
					Salary AVERAGE	#NAME?
					Salary AVERAGEA	#NAME?

The CEILING and ROUNDUP functions

Used to **round** numbers **up**. CEILING rounds up to the nearest **integer * multiple** of the specified ****factor**, while ROUNDUP rounds up to a certain **number of decimal places**.

=CEILING(value, [factor])

=ROUNDUP(value, [places])

- **value** – The value to be rounded off. Can contain the actual value or a cell reference.
- **factor/places** – The precision (number of decimal places) of the result.

* A **multiple** is a **product** of a given number with other numbers. For example, 6 is a multiple of 1 and 6 or 2 and 3.

** A **factor** **divides** a given number **without a remainder**. For example, 1, 2, 3, and 6 are factors of 6.

I2	fx	=CEILING(C2, 0.01)
K2	fx	=ROUNDUP(C2,0.01)

	C	I	J	K
1	Age (Year+Month)	CEILING		ROUNDUP
2	18.25	18.25		19

I2	fx	=CEILING(C2, 2)
K2	fx	=ROUNDUP(C2, 2)

	C	I	J	K
1	Age (Year+Month)	CEILING		ROUNDUP
2	18.25	20		18.25

The alternative CEILING functions

CEILING.PRECISE and **CEILING.MATH** functions round up to the nearest **integer** or **multiple** of the specified factor. **CEILING.MATH** also specifies if the number is rounded toward or away from zero.

I2	fx	=CEILING.PRECISE(C2, 0.01)
K2	fx	=CEILING.MATH(C2, 0.01, 0)

	C	I	J	K
1	Age (Year+Month)	CEILING.PRECISE	CEILING.MATH	
2	18.25	18.25	18.25	
3	20.67			

I2	fx	=CEILING.PRECISE(C2, 2)
K2	fx	=CEILING.MATH(C2, 2, -1)

	C	I	J	K
1	Age (Year+Month)	CEILING.PRECISE	CEILING.MATH	
2	-18.25	-18	-20	
3	20.67			

=CEILING.PRECISE(value, [factor])
=CEILING.MATH(value, [factor], [mode])

- **value** – The value to be rounded off. Can contain the actual value or a cell reference.
- **factor** – The precision (number of decimal places) of the result.
- **mode** – If the value is *negative*, it specifies the rounding direction. Zero or blank will round up towards zero. Otherwise, it's rounded away from zero.

The CEILING and ROUNDUP functions

Example use:

Round up the employee's **age** to the nearest **year**.

01. Round up the value of the first employee.

02. Replicate the formula to other employees by dragging down the fill handle.



Insight

This can be used to **determine** if any employees are expected to **retire within that year**.

Our **dataset shows** that all employees are between 18–36 years old, so **none will be retiring** any time soon.

D2			fx	=CEILING(C2,1)	
	A	B	C	D	E
1	Gender	JobTitle	Age (Year+Month)	Age (Years)	Education
2	Female	Graphic Designer	18.25	19	College
3	Male	Software Engineer	20.67		College
4	Female	Warehouse Associate	19.00		PhD

D20			fx	=CEILING(C20, 1)	
	A	B	C	D	E
1	Gender	JobTitle	Age (Year+Month)	Age (Years)	Education
2	Female	Graphic Designer	18.25	19	College
3	Male	Software Engineer	20.67	21	College
4	Female	Warehouse Associate	19.00	19	PhD
5	Male	Software Engineer	19.33	20	Masters
6	Male	Graphic Designer	25.50	26	Masters
7	Female	IT	20.00	20	PhD
8	Female	Graphic Designer	19.92	20	College
9	Male	Software Engineer	18.00	18	PhD
10	Female	Graphic Designer	32.58	33	High School
11	Female	Sales Associate	34.08	35	College
12	Male	Graphic Designer	24.00	24	PhD
13	Female	Driver	18.33	19	College

The CEILING and ROUNDUP functions

- 01. Will result in zero if the cell is **empty**.
- 02. Will return an **error** if the argument contains text or an error.

I2		<i>fx</i>	=CEILING(C2,2)							
		01.	<i>fx</i>	=ROUNDUP(C2,2)						
M2		<i>fx</i>	=CEILING.PRECISE(C2, 2)							
O2		<i>fx</i>	=CEILING.MATH(C2, 2,-1)							
		C		I	J	K	L	M	N	O
1		Age (Year+Month)	CEILING	ROUNDUP	CEILING.PRECISE	CEILING.MATH				
2			0	0	0	0				
3		20.67								
4		19.00								
5		19.33								
6		25.50								

I2	fx	=CEILING(C2,2)						
02.	fx	=ROUNDUP(C2,2)						
M2	fx	=CEILING.PRECISE(C2, 2)						
O2	fx	=CEILING.MATH(C2, 2,-1)						
	C	I	J	K	L	M	N	O
1	Age (Year+Month)	CEILING	ROUNDUP	CEILING.PRECISE	CEILING.MATH			
2	Analysis	#VALUE!	#VALUE!	#VALUE!	#VALUE!			
3	#NAME?	#NAME?	#NAME?	#NAME?	#NAME?			
4	19.00							
5	19.33							
6	25.50							

The FLOOR and ROUNDDOWN functions

FLOOR and ROUNDDOWN are used to **round** numbers **down**. FLOOR rounds a number down to the nearest **multiple** of the specified **factor**, while ROUNDDOWN rounds a number to a certain number of **decimal places**.

```
=FLOOR(value, [factor])
```

```
=ROUNDDOWN(value, [places])
```

- **value** – The value to round down.
- **factor/places** – The precision (number of decimal places) of the result. This argument is optional and is set to one by default (if not specified).

I2		<i>fx</i>	=FLOOR(E2,0.01)			
K2		<i>fx</i>	=ROUNDDOWN(E2,0.01)			
		E	I	J	K	M
1	BasePay	FLOOR	ROUNDDOWN			
2	42363.33	42363.33	42363			

I2	<i>fx</i>	=FLOOR(E2,2)			
K2	<i>fx</i>	=ROUNDDOWN(E2,2)			
	E	I	J	K	M
1	BasePay	FLOOR	ROUNDDOWN		
2	42363.33	42362	42363.33		

The alternative FLOOR functions

FLOOR.PRECISE and **FLOOR.MATH** round a number down to the nearest **integer** or **multiple** of the specified factor. **FLOOR.MATH** rounds down toward or away from zero.

I2	<i>fx</i>	=FLOOR.PRECISE(E2,0.01)
K2	<i>fx</i>	=FLOOR.MATH(E2,0.01,0)

	E	I	J	K	M
1	BasePay	FLOOR.PRECISE	FLOOR.MATH		
2	42363.33	42363.33	42363.33		
3	108476.67				

I2		=FLOOR.PRECISE(E2,2)
K2		=FLOOR.MATH(E2,2,-1)

	E	I	J	K	M
1	BasePay	FLOOR.PRECISE	FLOOR.MATH		
2	-42363.33	-42364	-42362		
3	108476.67				

=FLOOR.PRECISE(value, [factor])
=FLOOR.MATH(value, [factor], [mode])

- **value** – The value to round down.
- **factor** – The precision (number of decimal places) of the result. This argument is optional and is set to one by default (if not specified). It may not be equal to zero or be a cell reference to an empty cell.
- **mode** – If the value is negative, it specifies the rounding direction. Zero or blank will round away from zero. Otherwise, it's rounded towards zero.

The FLOOR and ROUNDDOWN functions

Example use:

Round down employee salaries to the nearest zero cents.

01. Round down the value of the first employee.

02. Replicate the formula to other employees by dragging down the fill handle.



Insight

It is a **common** practice to **round down** employees' **after-tax payment** to the nearest cent.

You could use this knowledge to investigate if the **BasePay** was captured **before or after taxation** to hypothesize why it was rounded down.

fx =FLOOR(E2,1)				
B	C	D	E	F
JobTitle	Age (Year+Month)	Education	BasePay	BasePay (Adjusted)
Graphic Designer	18.25	College	42363.33	42363.00
Software Engineer	20.67	College	108476.67	
Warehouse Associate	19.00	PhD	90208.25	
Software Engineer	19.33	Masters	108080.00	

fx =FLOOR(E20,1)				
B	C	D	E	F
JobTitle	Age (Year+Month)	Education	BasePay	BasePay (Adjusted)
Graphic Designer	18.25	College	42363.33	42363.00
Software Engineer	20.67	College	108476.67	108476.00
Warehouse Associate	19.00	PhD	90208.25	90208.00
Software Engineer	19.33	Masters	108080.00	108080.00
Graphic Designer	25.50	Masters	99464.80	99464.00
IT	20.00	PhD	70890.10	70890.00
Graphic Designer	19.92	College	67585.75	67585.00
Software Engineer	18.00	PhD	97523.33	97523.00
Graphic Designer	32.58	High School	112976.57	112976.00
Sales Associate	34.08	College	106524.40	106524.00
Graphic Designer	24.00	PhD	102261.01	102261.00
Driver	18.33	College	62759.99	62759.00
Financial Analyst	18.25	College	84007.80	84007.00

The FLOOR and ROUNDDOWN functions

01. Will result in zero if the cell is **empty**.
02. Will return an **error** if the cell contains text or an error.

I2	fx	=FLOOR(E2,2)
01.	fx	=ROUNDDOWN(E2,2)
M2	fx	=FLOOR.PRECISE(E2,2)
O2	fx	=FLOOR.MATH(E2,2,-1)

	E	I	J	K	L	M	N	O
1	BasePay	FLOOR	ROUNDDOWN	FLOOR.PRECISE	FLOOR.MATH			
2		0	0	0	0			
3	108476.67							
4	90208.25							
5	108080.00							
6	99464.80							

I2	fx	=FLOOR(E2,2)
02.	fx	=ROUNDDOWN(E2,2)
M2	fx	=FLOOR.PRECISE(E2,2)
O2	fx	=FLOOR.MATH(E2,2,-1)

	E	I	J	K	L	M	N	O
1	BasePay	FLOOR	ROUNDDOWN	FLOOR.PRECISE	FLOOR.MATH			
2	Analysis	#VALUE!	#VALUE!	#VALUE!	#VALUE!			
3	#NAME?	#NAME?	#NAME?	#NAME?	#NAME?			
4	90208.25							
5	108080.00							
6	99464.80							

The MOD function

MOD is short for modulo or **modulus**. It divides a number and then gives the **remainder** as an answer.

=MOD(dividend, divisor)

- **dividend** – The number being divided.
- **divisor** – The number by which you want to divide the dividend argument. The divisor may not be equal to zero or be a cell reference to an empty cell.

For example, **5 mod 3 = 2**

5 is the dividend, 3 is the divisor, and 2 is the modulo.

/x =MOD(E2,100)				
B	C	D	E	F
Job Title	Age (Year+Month)	Education	BasePay	BasePay Modulus
Graphic Designer	18.25	College	42363.33	63.33
Software Engineer	20.67	College	108476.67	
Warehouse Associate	19.00	PhD	90208.25	
Software Engineer	19.33	Masters	108080.00	
Graphic Designer	25.50	Masters	99464.80	
T	20.00	PhD	70890.10	
Graphic Designer	19.92	College	67585.75	
Software Engineer	18.00	PhD	97523.33	
Graphic Designer	32.58	High School	112976.57	
Sales Associate	34.08	College	106524.40	
Graphic Designer	24.00	PhD	102261.01	
Driver	18.33	College	62759.99	
Financial Analyst	18.25	College	84007.80	
Warehouse Associate	29.67	Masters	86220.33	

The MOD function

fx =MOD(E2,100)				
B	C	D	E	F
JobTitle	Age (Year+Month)	Education	BasePay	BasePay Modulus
Graphic Designer	18.25	College	42363.33	63.33
Software Engineer	20.67	College	108476.67	
Warehouse Associate	19.00	PhD	90208.25	

fx =MOD(E20,100)				
B	C	D	E	F
JobTitle	Age (Year+Month)	Education	BasePay	BasePay Modulus
Graphic Designer	18.25	College	42363.33	63.33
Software Engineer	20.67	College	108476.67	76.67
Warehouse Associate	19.00	PhD	90208.25	8.25
Software Engineer	19.33	Masters	108080.00	80.00
Graphic Designer	25.50	Masters	99464.80	64.80
IT	20.00	PhD	70890.10	90.10
Graphic Designer	19.92	College	67585.75	85.75
Software Engineer	18.00	PhD	97523.33	23.33
Graphic Designer	32.58	High School	112976.57	76.57
Sales Associate	34.08	College	106524.40	24.40
Graphic Designer	24.00	PhD	102261.01	61.01
Driver	18.33	College	62759.99	59.99
Financial Analyst	18.25	College	84007.80	7.80

Example use:

Identify **salaries** that are **not multiples of 100**.

01. Calculate the remainder of the first employee's salary when divided by 100.
02. Replicate the formula to other employees by dragging down the fill handle.



Insight

Our dataset fully consists of data that are **not multiples of 100**.

This could suggest that employees are **paid on an hourly basis**—a hypothesis worth checking out!

The MOD function

01. Will result in zero if the cell is **empty**.
02. Will return an **error** if the argument contains text or an error.

fx | =MOD(E2,100)

01.

	C	D	E	F
JobTitle	Age (Year+Month)	Education	BasePay	BasePay Modulus
Graphic Designer	18.25	College		0.00
Software Engineer	20.67	College	108476.67	76.67
Warehouse Associate	19.00	PhD	90208.25	8.25
Software Engineer	19.33	Masters	108080.00	80.00
Graphic Designer	25.50	Masters	99464.80	64.80
IT	20.00	PhD	70890.10	90.10
Graphic Designer	19.92	College	67585.75	85.75
Software Engineer	18.00	PhD	97523.33	23.33

fx | =MOD(E3,100)

02.

	C	D	E	F
JobTitle	Age (Year+Month)	Education	BasePay	BasePay Modulus
Graphic Designer	18.25	College	Analysis	#VALUE!
Software Engineer	20.67	College	#NAME?	#NAME?
Warehouse Associate	19.00	PhD	90208.25	8.25
Software Engineer	19.33	Masters	108080.00	80.00
Graphic Designer	25.50	Masters	99464.80	64.80
IT	20.00	PhD	70890.10	90.10
Graphic Designer	19.92	College	67585.75	85.75
Software Engineer	18.00	PhD	97523.33	23.33

The Sqrt and Power/Pow functions

Sqrt is used to calculate the square root of a given number, while Pow, which is equivalent to Power, is used to calculate the value of a number (base) raised to a certain exponent.

=SQRT(value)

- **value** – The number whose square root we want to calculate.

For example, $\sqrt{4}$ will be entered as **=SQRT(4)**

=POW(base, exponent)
=POWER(base, exponent)

- **base** – Any real number.
- **exponent** – The number of times the base will be multiplied by itself.

For example, 2^3 will be entered as **=POW/POWER(2,3)**

The Sqrt and Power/Pow functions

This screenshot shows the formula bar and the first two rows of data. In the formula bar, cell C2 contains the formula `=POW(A2,3)` and cell E2 contains `=SQRT(A2)`. The table below shows the data for these cells.

	A	B	C	D	E	F
1	Number		Power		Square Root	
2	2		8		1	
3	4					
4	9					

Example use:

When working with **mathematical equations**.

This screenshot shows the full data table. The formulas for the 'Power' and 'Square Root' columns are highlighted with a red box, showing how they are applied to the 'Number' column.

	A	B	C	D	E	F
1	Number		Power		Square Root	
2	2		8		1	
3	4		64		2	
4	9		729		3	
5	16		4096		4	
6	25		15625		5	
7	36		46656		6	
8	49		117649		7	
9	64		262144		8	
10	81		531441		9	
11	100		1000000		10	

01. Calculate the square root and power of the first number.
02. Replicate the formula to other rows by dragging down the fill handle.

The SQRT and POWER/POW functions

01. SQRT only works with **positive numbers** or **zero**, while POW (or POWER) works with **positive** and **negative** numbers.
02. SQRT and POW (or POWER) will result in an **error** if the argument is **text** or an **error**.

01.

C3

fx

=POW(A3,3)

fx

=SQRT(A3)

	A	B	C	D	E	F	G
1	Number		Power		Square Root		
2	0		0		0		
3	-4		-64		#NUM!		
4	9		729		3		
5	16		4096		4		
6	25		15625		5		
7	36		46656		6		
8	49		2401		7		

C2

fx

=POW(A2,3)

02.

fx

=SQRT(A2)

	A	B	C	D	E	F	G
1	Number		Power		Square Root		
2	Analysis		#VALUE!		#VALUE!		
3	#NAME?		#NAME?		#NAME?		
4	9		729		3		
5	16		4096		4		
6	25		15625		5		
7	36		46656		6		
8	49		117649		7		